Optimized Field-Weakening Operation of PMSM Modulated Model Predictive Control Using Predictive Current Error Margin

Qinghua Dong, Bo Wang, *Member, IEEE*, Liqun Xia, Yong Yu, Minghe Tian, and Dianguo Xu, *Fellow, IEEE*

***Abstract*—Considering the intrinsic defects of finite control set MPC (FCS-MPC), e.g., torque ripple and various harmonic frequencies, the modulated model predictive control (M2PC) has been regarded as a promising approach to achieve high performance of permanent magnet synchronous motor (PMSM) drives. However, the M2PC still faces non-minimum tracking error and unreasonable trajectory settings when rated-speed operation extends to the field-weakening (FW) region since the intervention of multiple constraints of voltage and current has to be taken into account. To address this problem, an M2PC-based scheme is proposed to realize the field-weakening operation with minimum tracking error and reasonable trajectory setting. Specifically, the equal proportion algorithm is replaced by an algebraical way for minimum tracking error in the over-modulation region; the predictive current error margin is used to preset the current trajectory in FW operation. Besides, the pre-triggering method is applied to reduce the control delay and eliminate the cumulative error, which improves the tolerance of parameter perturbation. Comparison results conducted on a Zynq7020-based PMSM setup validate the superiority of the proposed method regarding the capability of high-speed operation and high-dynamic performance.**

***Index Terms*—Permanent magnet synchronous motors, modulated model predictive control, field-weakening operation, predictive current error margin.**

I. Introduction

PERMANENT magnet synchronous motors (PMSMs) have been widely applied in industrial production owing to their high power density, high efficiency, and low maintenance cost [1], [2], [3]. In some specific applications, such as centrifuges, high-precision CNC machine tools, and electro-hydrostatic actuators, excellent control performances (e.g., fast dynamic response, accurate current control, and wide operation range) are required for the drive systems [4], [5].

Manuscript received 4 April 2023; revised 7 June 2023; accepted 30 July 2023. Date of publication 15 August 2023; date of current version 21 February 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 52177030 and in part by the Natural Science Foundation of Heilongjiang Province of China under Grant YQ2020E018. Paper no. TEC-00376-2023. *(Corresponding authors: Bo Wang; Yong Yu.)*

The authors are with the School of Electrical Engineering and Automation, Harbin Institute of Technology, Harbin 150001, China (e-mail: dqh_hit@163.com; wangbohit@hit.edu.cn; xialiqun618fcd@163.com; yuyong@hit.edu.cn; minghe_tian@163.com; xudiang@hit.edu.cn).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TEC.2023.3305303.

Digital Object Identifier 10.1109/TEC.2023.3305303

Finite control set model predictive control (FCS-MPC) is famous for its high dynamic response in motor control [6], [7], [8], [9], [10]. However, the inherent principle of FCS-MPC, such as the single voltage vector and the unfixed switching frequency, will cause a large torque ripple and the spread current harmonic spectrum [11], etc. Hence, it is difficult for FCS-MPC to ensure the high-precision cutting in CNC machine tools and the high-precision positioning in electro-hydrostatic actuators. Besides, the spread current spectrum also aggravates the difficulty of designing output filters, especially for high-speed motors with low inductance. To solve the issue mentioned above, modulated MPC (M2PC) has been introduced to achieve the small torque ripple and the fixed switching frequency, and it has been considered a promising method for high-performance motor control [11], [12].

Recently, various M2PC-based approaches have been studied to improve the accuracy of the required duty cycle by the predictive current error (PCE), where PCE is the difference between the reference value and the predictive value [13], [14], [15], [16], [17], [18]. The method proposed in [15] is to solve the effective duty cycle by the ratio of adjacent voltage cost function, simplifying the duty cycle calculation. In [16], the duty cycle is calculated by minimizing the root-mean-square (RMS) of PCE, realizing the acquisition of a multi-level duty cycle and application for the 3L-NPC converters. [17] obtains the accurate duty cycle according to the synthesis of PCE vector. As the motor speed increases, the required voltage will transit from the linear region to the nonlinear region under the intervention of inverter constraint. Although the above method realizes the duty cycle calculation in the linear modulation region, it is not entirely suitable for the nonlinear modulation region. In [18], the duty cycle calculation is optimized by the geometric method, which increases the computational complexity due to the involvement of trigonometric functions.

Further, when a higher operating speed is required, the motor has to enter the field-weakening (FW) region. In the FW region, the establishment of an operation trajectory regulation mechanism is the prerequisite to avoid conflicts between motor operation and maximum voltage and current limitations [19], [20], [21], [22], [23]. From the viewpoint of MPC framework, the FW algorithms can be classified into two main categories: prior-based, i.e., directly designing the cost function with

0885-8969 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

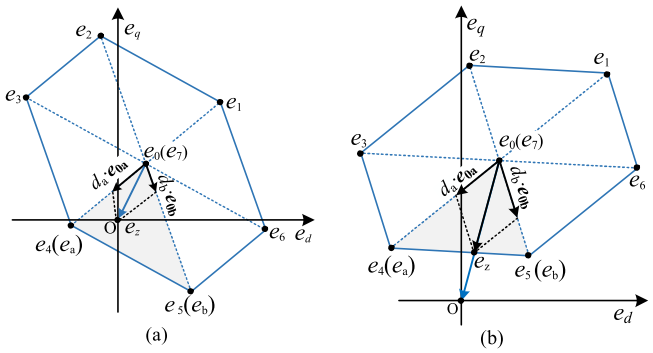


Fig. 1. Diagram of PCE in e_d-e_q plane. (a) Linear modulation region. (b) Over-modulation region.

multiple constraints [24], [25], [26]; or posterior-based, i.e., adjusting preset through involved multiple constraints [27], [28], [29], [30].

In [25], the voltage and current constraints are introduced into the cost function by implicit formulation to ensure that the operation trajectory migrates within the allowed range. In [26], speed control is added to the cost function except for the control objective and constraints of [25], which realizes speed predictive control in the full-speed range. Although these methods focusing on cost function can prescribe the operation trajectory in the FW region, they will exacerbate the complexity of the implementation since the control dimension expands from two-dimensional to multi-dimensional due to the intervention of multiple constraints. In addition, it is difficult to directly integrate these methods into M2PC that requires vector synthesis due to the increase of control dimension.

In [27], an explicit MPC algorithm is applied to guide FW operation for PMSM drives, in which the load capacity will be limited due to the lack of a feedback mechanism. For the FW regulation in [28] and [29], the reference torque is directly calculated by the voltage, current, and flux linkage vector angle limits. These methods can optimize torque distribution to some extent, while the reference flux is still set according to the open-loop mode. [30] presents an FW strategy based on MPC with a simple cost function, which generates the flux compensation with voltage constraint enforced. For the line of approach that adjusts the preset through multiple constraints, its superiority lies in that the flux and torque are calculated by the interaction of multiple constraints, and thereby the cost function can be freed up for reference tracking. However, since there is no feedback mechanism to calibrate the reference, accurate distribution of flux and torque still cannot be guaranteed, which possibly results in deep saturation of the inverter and insufficient transient performance. Therefore, it is significant for the M2PC-based FW strategy to optimize the FW trajectory achievement.

To address the above problems, an M2PC-based field-weakening method is proposed to achieve field-weakening operation with the minimum tracking error and a reasonable operating trajectory setting. The specific contributions are as follows. (1) The algebraical algorithm is designed to replace the equal proportion algorithm, which is capable of effectively reducing the PCE and improving the tracking capacity in the over-modulation region. (2) The PCE margin is proposed to characterize the saturation of the inverter, and it also serves as the feedback for the accurate distribution of flux linkage and torque. (3) Moreover, the pre-triggering method is adopted to effectively reduce the control delay and eliminate the cumulative error of traditional delay compensation.

The rest of this article is organized as follows: the high-speed operation of the conventional M2PC is reviewed in Section II. The proposed control scheme for enhanced high-speed operation is presented in Section III, followed by the experimental verification in Section IV.

II. High-Speed Operation of the Conventional M2PC

A. Basic Principle of the Conventional M2PC

The stator voltage equations of PMSM in d-q synchronous rotating frame can be described as

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = \begin{bmatrix} R_{\rm s} & -\omega_{\rm e} L_q \\ \omega_e L_d & R_{\rm s} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} \frac{{\rm d}i_d}{{\rm dt}} \\ \frac{{\rm d}i_q}{{\rm dt}} \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_{\rm e}\psi_f \end{bmatrix} \tag{1}$$

where u_d, u_q, i_d, i_q, L_d, and L_q are the voltage, current, and inductance in dq-axis, respectively. $R_{\rm s}$ is the stator resistance, d is the differential operator, $\omega_{\rm e}$ is the electrical angular speed, ψ_f is the permanent magnet flux linkage.

The predictive current equation of M2PC can be obtained by the forward Euler discretization of (1), as

$$\begin{bmatrix} i_d(k+1) \\ i_q(k+1) \end{bmatrix} = \begin{bmatrix} 1-\frac{RT_{\rm s}}{L_d} & \frac{\omega_{\rm e}(k) L_q T_{\rm s}}{L_d} \\ -\frac{\omega_{\rm e}(k) L_d T_s}{L_q} & 1-\frac{RT_{\rm s}}{L_q} \end{bmatrix} \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix} + \begin{bmatrix} \frac{T_{\rm s}}{L_d} & 0 \\ 0 & \frac{T_{\rm s}}{L_q} \end{bmatrix} \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{-\psi_f T_{\rm s}}{L_q}\omega_{\rm e}(k) \end{bmatrix} \tag{2}$$

where $T_{\rm s}$ is the sampling period, k is the discrete-time point. The steps of M2PC are summarized as follows.

1) *Step 1:* Calculate the PCE for each available voltage. For the application of PMSM driver with current control, the PCE function is expressed as

$$\begin{bmatrix} e_{jd} \\ e_{jq} \end{bmatrix} = \begin{bmatrix} i_d^*(k+1) - i_d(k+1)|u_{jd} \\ i_q^*(k+1) - i_q(k+1)|u_{jq} \end{bmatrix} \tag{3}$$

where e_d, e_q are PCE of the d-axis and q-axis, subscript j is the label of discrete voltage in the two-level inverter, and $j \in \{0, 1, 2 \ldots 7\}$, i_d^* and i_q^* are the reference currents of d-axis and q-axis, respectively. The point $e_j = (e_{jd}, e_{jq})$ is marked in the time-varying e_d-e_q coordinate, as shown in Fig. 1.

2) *Step 2:* Select the required sector by the vectors cross product and dot product, as

$$\mathrm{m}(e_{\rm a}, e_{\rm b}) = \{((e_{\rm a} - e_0) \times (-e_0)) \cdot ((e_{\rm b} - e_0) \times (-e_0)) \le 0\}$$

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

$$\wedge \{(e_{\rm a} - e_0) \centerdot (-e_0) > 0\} \wedge \{(e_{\rm b} - e_0) \centerdot (-e_0) > 0\} \tag{4}$$

where m(·) is the boolean function, and subscripts a and b are two adjacent vectors in the selected sector. In Fig. 1, $e_{\rm a}$ and $e_{\rm b}$ refer to e_4 and e_5.

3) *Step 3:* Calculate the duty cycle by steering PCE to the origin *O* in the line modulation, as

$$\begin{aligned}(e_{ad} - e_{0d})\, d_{\rm a} + (e_{{\rm b}d} - e_{0d})\, d_{\rm b} + e_{0d} = 0\\ (e_{aq} - e_{0q})\, d_{\rm a} + (e_{{\rm b}q} - e_{0q})\, d_{\rm b} + e_{0q} = 0\end{aligned} \tag{5}$$

where $d_{\rm a}$ and $d_{\rm b}$ are the duty cycle of PCE vector. Fig. 1(a) shows the diagram of PCE in the linear modulation region.

4) *Step 4:* Shorten the duty cycle with equal proportion if it is in the over-modulation region, as

$$\begin{cases} d_{\rm a} = d_{\rm a}/(d_{\rm a} + d_{\rm b}) \\ d_{\rm b} = d_{\rm b}/(d_{\rm a} + d_{\rm b}) \end{cases} \quad \text{if} \quad d_{\rm a} + d_{\rm b} > 1. \tag{6}$$

Fig. 1(b) shows the diagram of PCE in the over-modulation region.

5) *Step 5:* Select the corresponding voltage vector and duty cycle, and generate $T_{\rm on}$ (switch-on time) by the seven-segment PWM generation rules.

B. Problem of Filed-Weakening Operation on M2PC

For the application of motor control, with the increase of motor speed, the operation will inevitably enter the over-modulation region, as shown in Fig. 1(b). During the transient process, the origin O will fall outside the PCE boundary due to the inverter saturation, leading to the failure of the predictive current tracking. In conventional M2PC, the duty cycle is adjusted by the equal proportion method (6) in the over-modulation region. From Fig. 1(b), the point $e_{\rm z}$, representing the end of the selected PCE vector, is not the closest point to the origin O within the boundary of PCE, resulting in the non-minimum PCE amplitude. Furthermore, the tracking error caused by non-minimum PCE will become more serious with the deepening modulation, which will restrict the dynamic performance improvement.

To further extend the speed, the establishment of FW mechanism is a premise. However, it is difficult for the existing MPC-based FW methods to achieve high-performance FW operation in M2PC. In the method of the cost function with multiple constraints, the cost function is composed of multiple sub-functions. It means that the control dimension increases from the two dimensions (i.e., torque and flux) to multi-dimensions (i.e., torque, flux, and constraints). However, the M2PC mentioned above is not suitable for vector synthesis in multi-dimensions due to the increase of control dimensions. Besides, the cost function with multi-dimensions will consume a lot of computing resources. Another method is to adjust the preset through multiple constraints, where the FW trajectory is directly guided by the open-loop model. Owing to the lack of a feedback mechanism, it is difficult to obtain the accurate distribution of flux and torque according to the actual condition.

Moreover, in the digital control system, the control delay is one of the essential factors affecting performance. In the traditional delay compensation method, the cumulative error is an unavoidable problem since the current of the next moment is prepaid in the present moment, and it becomes more pronounced when the parameters are varied.

Therefore, to realize the improvement for field-weakening operation, this article optimizes the M2PC-based control algorithm from three aspects, including tracking error reduction, FW trajectory guidance, and control delay circumvent.

III. Proposed M2PC-Based High-Speed Control Scheme

In this section, an enhanced M2PC-based field-weakening control scheme is proposed, as shown in Fig. 2. The reference *q*-axis current is calculated from the speed regulator and current limitation. The reference *d*-axis current is obtained from the speed regulator, maximum torque per ampere (MTPA), and field-weakening current. Then, the duty cycle is calculated by the proposed FW-M2PC method according to the reference *dq*-axis current, the feedback *dq*-axis current, rotor speed, rotor position, and dc-bus voltage.

In the proposed method, the conventional M2PC is redesigned by the PCE cost function, calculating the duty cycle through algebraical algorithms, as shown in Part A. The PCE margin is introduced to monitor the saturation of the inverter output voltage, and an FW scheme using the PCE margin as feedback is designed to guide the operation trajectory and adjust the distribution of the *dq*-axis currents, which will be discussed in Part B. In Part C, the pre-triggering method is applied to reduce the control delay and eliminate the cumulative error.

A. Algebraical M2PC Method With PCE Minimization

In the current control of M2PC, tracking error minimization is equivalent to PCE minimization. Fig. 3 is the diagram of the minimum PCE in the over-modulation region. From Fig. 3, the magnitude of PCE is equivalent to the distance from the origin *O* to the PCE boundary. The orange segments in Fig. 4 are the sets of the distance. Thus, to obtain the duty cycle for the shortest distance, the distance cost function is expressed as

$$\begin{aligned} g &= \sqrt{e_{{\rm z}d}^2 + e_{{\rm z}q}^2} \\ e_{{\rm z}d} &= (e_{ad} - e_{0d})\, d_{\rm a} + (e_{{\rm b}d} - e_{0d})\, d_{\rm b} + e_{0d} \\ e_{{\rm z}q} &= (e_{aq} - e_{0q})\, d_{\rm a} + (e_{{\rm b}q} - e_{0q})\, d_{\rm b} + e_{0q} \end{aligned} \tag{7}$$

where $e_{{\rm z}\,d}$ and $e_{{\rm z}\,q}$ are the coordinates of the point $e_{\rm z}$ in e_d-axis and e_q-axis frame, respectively. Point $e_{\rm z}$ is the end of the selected PCE vector. The distance cost function is also called the PCE cost function due to the principle of equivalence.

As can be seen from Fig. 3, the moving range of point $e_{\rm z}$ is between point $e_{\rm a}$ and point $e_{\rm b}$, as the red segment shows. When the point $e_{\rm z}$ coincides with the point $e_{\rm a}$, $d_{\rm a}$ is maximum ($d_{\rm a} = 1$). When the point $e_{\rm z}$ coincides with the point $e_{\rm b}$, $d_{\rm b}$ is maximum ($d_{\rm b} = 1$). So, the constraints of the solution of (7) can be written as

$$d_{\rm a} + d_{\rm b} = 1, \quad d_{\rm a} \in [0, 1]. \tag{8}$$

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

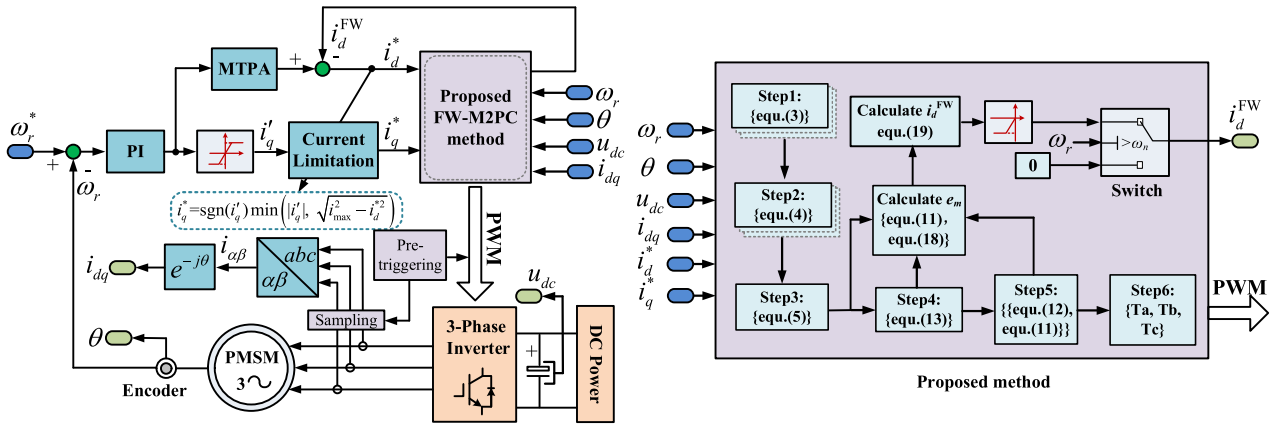


Fig. 2. Control block diagram of the proposed method.

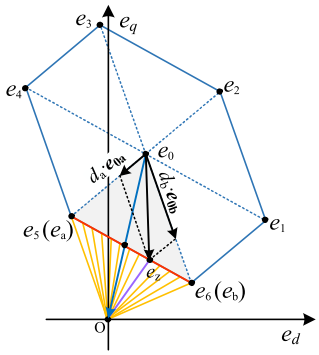


Fig. 3. Diagram of the minimum PCE in the over-modulation region.

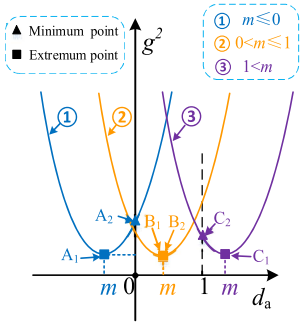


Fig. 4. Diagram of d_a and g^2 in different m.

Substituting (7) into (8), the square of PCE cost function is written as

$$\begin{aligned} g^2 = & \left[(e_{ad}-e_{bd})^2+(e_{aq}-e_{bq})^2\right] d_a^2 \\ & + 2\left[e_{bd}\left(e_{ad}-e_{bd}\right)\right. \\ & \left. + e_{bq}(e_{aq}-e_{bq})\right] d_a + e_{bd}^2 + e_{bq}^2 \quad d_a \in [0,1]. \end{aligned} \tag{9}$$

From (9), the square of PCE cost function is a parabola with variable d_a. The solution of the shortest distance can be transformed into solving the minimum value of the parabola with constraints. Thus, the extremum of (9) can be derived, as

$$\frac{\mathrm{d}g^2}{\mathrm{d}d_a} = 0 \quad \Rightarrow d_a = m = -\frac{e_{bd}(e_{ad}-e_{bd})+e_{bq}(e_{aq}-e_{bq})}{(e_{ad}-e_{bd})^2+(e_{aq}-e_{bq})^2}. \tag{10}$$

Suppose the coordinate of the extremum point is (m, g_{ext}^2). Due to the variable $d_a \in [0, 1]$, the extremum (g_{ext}^2) is not necessarily the minimum of g^2 (g_{min}^2) under the different m. Fig. 4 shows the diagram of d_a and g^2 in different m. In case ① (i.e., $m \leq 0$), point A_2 is the minimum point under constraint conditions. Similarly, points B_2 and C_2 are the minimum points of ② and ③, respectively. Thus, the value of g_{min}^2 can be solved as

$$g_{min}^2 = \begin{cases} e_{bd}^2 + e_{bq}^2 & m < 0 \\ \frac{(e_{ad}e_{bq}-e_{bd}e_{aq})^2}{(e_{ad}-e_{bd})^2+(e_{aq}-e_{bq})^2} & 0 \leq m < 1 \\ e_{ad}^2 + e_{aq}^2 & m \geq 1 \end{cases}. \tag{11}$$

Since the current constraint is not considered in Fig. 3, it is not possible to determine whether points A_2 and C_2 satisfy the current constraint. According to the derivation above, A_2 and C_2 are the PCE minimum points, and thus the feedback current can track the reference value as fast as possible. Therefore, in the proposed method, the feedback value is indirectly constrained in the form of a constrained reference value, as shown in the current limitation module of Fig. 2.

According to (8) and (11), the duty cycle at the shortest distance is described as

$$\begin{cases} d_a = 0,\ d_b = 1 & m < 0 \\ d_a = m,\ d_b = 1-m & 0 \leq m < 1 \\ d_a = 1,\ d_b = 0 & m \geq 1 \end{cases}. \tag{12}$$

In broad terms, the modified M2PC works with six steps, as shown in Fig. 5. The judgment of *step*4 is described as

$$d_a \in [0,1],\quad d_b \in [0,1],\quad d_a + d_b \in [0,1]. \tag{13}$$

Notice that the modified M2PC method sets up a PCE cost function to select the duty cycle, achieving the minimum PCE. The whole solution process only involves simple algebraic calculation, and thus it is easy to implement in a microprocessor.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

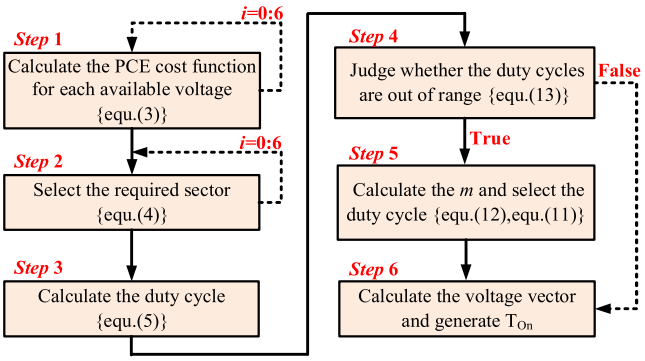


Fig. 5. Six steps of the algebraical M2PC.

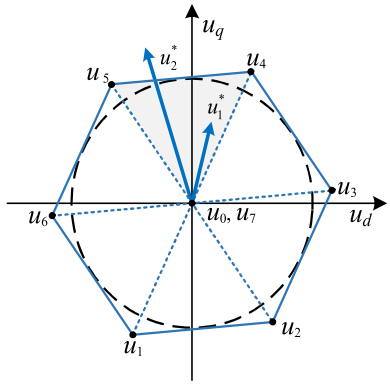


Fig. 6. Voltage vectors in the plane of u_d-u_q.

From Fig. 3 and (7), it can be seen that the PCE cost function g de facto represents the amplitude of PCE, as well as the saturation depth of the inverter output in the over-modulation region. Thus, g is considered as the reference to adjust the FW trajectory, which will be discussed in the next part.

B. FW Trajectory Guidance Using PCE Margin

In the FW operation of PMSM, accurate trajectory guidance (i.e., accurate current distribution) is of particular importance to enhance the dynamic performance and load capacity. In this region, the motor is mainly constrained by the maximum voltage and the maximum current, as

$$u_d^2 + u_q^2 \le u_{\max}^2 \tag{14}$$

$$i_d^2 + i_q^2 \le i_{\max}^2. \tag{15}$$

Since the dq-axis currents are the control object in this system, the current constraint can be directly imposed on the reference current. As the current limitation shows in Fig. 2, the reference of q-axis current is constrained by the maximum current and the reference of d-axis current. Thus, only the voltage constraint needs to be considered. Substituting (14) into (2), the voltage constraint can be converted to predictive current constraint in k instant, as

$$\left[\left(R_s - \frac{L_d}{T_s}\right) i_d(k) - \omega_e(k) L_q i_q(k) + \frac{L_d}{T_s} i_d(k+1)\right]^2 + \left[\left(R_s - \frac{L_q}{T_s}\right) i_q(k) + \omega_e(k) L_d i_d(k) + \omega_e(k)\psi_f + \frac{L_q}{T_s} i_q(k+1)\right]^2 \le u_{\max}^2. \tag{16}$$

According to (3) and (16), the predictive current constraint can be converted to the PCE constraint, as

$$\frac{L_d^2}{T_s^2}(e_d - e_{0d})^2 + \frac{L_q^2}{T_s^2}(e_q - e_{0q})^2 \le u_{\max}^2 \tag{17}$$

where

$$e_{0d} = \frac{T_s}{L_d}\left(\left(R_s - \frac{L_d}{T_s}\right) i_d(k) - \omega_e(k) L_q i_q(k) + \frac{L_d}{T_s} i_d^*(k+1)\right)$$

$$e_{0q} = \frac{T_s}{L_q}\left(\left(R_s - \frac{L_q}{T_s}\right) i_q(k) + \omega_e(k) L_d i_d(k) + \omega_e(k)\psi_f + \frac{L_q}{T_s} i_q^*(k+1)\right)$$

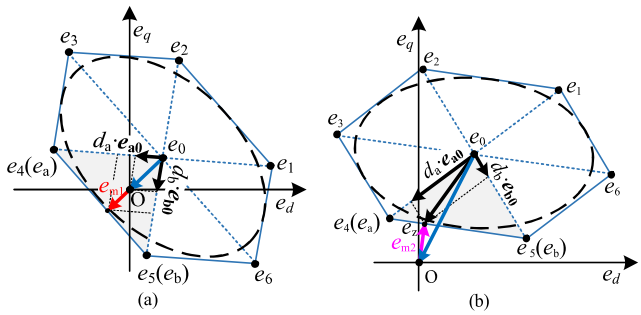


Fig. 7. PCE margin in the plane of e_d-e_q. (a) *step 4* = false. (b) *step 4* = true.

From (17), if $u_{\max}$ is a constant, the range of the PCE constraint would be an ellipse in $k+1$ instant ($L_d \ne L_q$). The center point of the ellipse is (e_{0d}, e_{0q}), and its coordinate depends on the state variables in k instant and the references in $k+1$ instant. According to the above deduction, the voltage constraint can be mapped into the constraint of PCE, as the black dotted lines shown in Figs. 6 and 7.

In fact, the limitation boundary of PCE is an irregular hexagon in the algebraic M2PC, as shown in Fig. 3. Based on the mapping of voltage and PCE, the maximum utilization range of voltage is not circular but a regular hexagon, and thus $u_{\max}$ is not a constant but variable in the algebraic M2PC. Thus, the boundary of PCE constraint is an irregular hexagon rather than an ellipse, as shown in Fig. 7.

In FW control, the intersection of voltage and current constraints constitutes the desired operating trajectory; that is, the output voltage of the inverter is maintained at quasi-saturation. According to the mapping between the PCE constraint and the voltage constraint, a concept of PCE margin (e_m) is introduced to detect the saturation depth of the inverter accurately. (18) is the

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

formula of e_{m}, and *step* 4 mentioned in part A is the judgment of whether the reference PCE vector ($\mathbf{e_0o}$) is beyond the boundary. When $\mathbf{e_0o}$ is within the boundary, the inverter has enough margin to follow the reference of *dq*-axis currents in $k+1$ instant. In this case, e_{m} is defined as the extension of $\mathbf{e_0o}$ to the boundary, as the solid red line (e_{m1}) shows in Fig. 7(a). The amplitude of e_{m1} is obtained by the principle of triangle similarity, and the sign is set to positive. If $\mathbf{e_0o}$ exceeds the boundary, the e_{m} is defined as the error between $\mathbf{e_0o}$ and the actual PCE vector ($\mathbf{e_{0z}}$). The solid pink line in Fig. 7(b) is the PCE margin (e_{m2}). The amplitude of e_{m2} is the value of PCE cost function g mentioned in part A, and the sign is set to negative. Therefore, e_{m} can be described as

$$e_{\mathrm{m}} = \begin{cases} \frac{d_0}{d_{\mathrm{a}}+d_{\mathrm{b}}}\sqrt{e_{0d}^2+e_{0q}^2} & step\ 4 = \text{false} \\ -g_{\min} & step\ 4 = \text{true} \end{cases}. \tag{18}$$

According to the above analysis, if the PCE margin is controlled to zero, the inverter will be in quasi-saturation. It means that the PCE margin can be used as feedback to adjust the current distribution. Meanwhile, to complete the feedback mechanism, a PI regulator is applied as the actuator of this mechanism, and thus the FW current can be written as

$$i_d^{\mathrm{FW}} = \lim\left[\left(k_{\mathrm{p}} + k_{\mathrm{i}}/s\right) e_{\mathrm{m}}\right] \tag{19}$$

where i_d^{FW} is the FW current, lim[] is the limiting value of PI controller, k_{p} and k_{i} are the parameters of PI controller. The upper and lower limits are set to zero and maximum negative current, respectively. Increasing k_{p} improves the response of the system but causes overshooting, and increasing k_{i} reduces the steady-state error of the system but causes oscillation. The trial-and-error method is the solution to obtain suitable PI parameters. Besides, a switch is needed to prevent the incorrect current regulation since $\mathbf{e_0o}$ will rapidly exceed the boundary in motor start-up due to the high gain of M2PC. The feedback mechanism is activated only when the motor speed exceeds the rated speed.

C. Pre-Triggering for Control Delay Reduction

In the M2PC, one-delay compensation is a classical approach to compensate for the control delay, but it also introduces cumulative errors in the case of parameter perturbation. In the proposed method, pre-triggering sampling is applied to reduce the control delay and eliminate the cumulative error.

The essence of the one-delay compensation method [31] is a two-horizon prediction method. Fig. 8 presents the diagram of the two-horizon prediction. For clarity, the one-delay compensation method can be described as

$$\boldsymbol{i}(k+1) = \mathbf{A}\boldsymbol{i}(k) + \mathbf{B}\boldsymbol{u}(k) + \mathbf{D} \tag{20}$$

$$\boldsymbol{i}(k+2) = \mathbf{A}\boldsymbol{i}(k+1) + \mathbf{B}\boldsymbol{u}(k+1) + \mathbf{D} \tag{21}$$

where

$$\boldsymbol{i}(k) = \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix}, \ \mathbf{A} = \begin{bmatrix} 1-\frac{R_{\mathrm{s}}T_{\mathrm{s}}}{L_d} & \frac{\omega_{\mathrm{e}}(k)L_qT_{\mathrm{s}}}{L_d} \\ \frac{-\omega_{\mathrm{e}}(k)L_dT_{\mathrm{s}}}{L_q} & 1-\frac{R_{\mathrm{s}}T_{\mathrm{s}}}{L_q} \end{bmatrix},$$

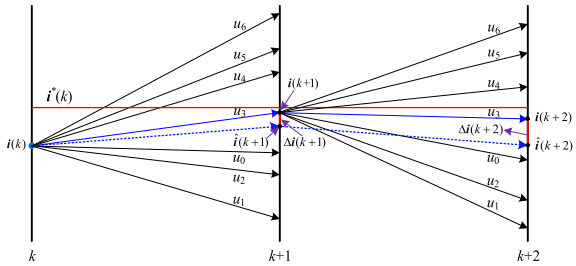


Fig. 8. Diagram of two-horizon prediction.

$$\boldsymbol{u}(k) = \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} \frac{T_{\mathrm{s}}}{L_d} & 0 \\ 0 & \frac{T_{\mathrm{s}}}{L_q} \end{bmatrix}, \ \mathbf{D} = \begin{bmatrix} 0 \\ \frac{-T_{\mathrm{s}}}{L_q}\psi_f\omega_{\mathrm{e}}(k) \end{bmatrix}.$$

From Fig. 8, the current at $k+2$ is predicted based on the current at $k+1$ which predicted from the sampling current at k. Since the current at $k+1$ is not the sampling current, it will inevitably introduce cumulative errors when the parameters are perturbed during the two-horizon prediction.

The process of generating the cumulative error is as follows. Predictive (20) and (21) can be rewritten as

$$\hat{\boldsymbol{i}}(k+1) = \bar{\mathbf{A}}\boldsymbol{i}(k) + \bar{\mathbf{B}}\boldsymbol{u}(k) + \bar{\mathbf{D}} \tag{22}$$

$$\hat{\boldsymbol{i}}(k+2) = \bar{\mathbf{A}}\hat{\boldsymbol{i}}(k+1) + \bar{\mathbf{B}}\boldsymbol{u}(k+1) + \bar{\mathbf{D}} \tag{23}$$

where

$$\bar{\mathbf{A}} = \begin{bmatrix} 1-\frac{\bar{R}_{\mathrm{s}}T_{\mathrm{s}}}{\bar{L}_d} & \frac{\omega_{\mathrm{e}}(k)\bar{L}_qT_{\mathrm{s}}}{\bar{L}_d} \\ -\frac{\omega_{\mathrm{e}}(k)\bar{L}_dT_{\mathrm{s}}}{\bar{L}_q} & 1-\frac{\bar{R}T_{\mathrm{s}}}{\bar{L}_q} \end{bmatrix}, \ \bar{\mathbf{B}} = \begin{bmatrix} \frac{T_{\mathrm{s}}}{\bar{L}_d} & 0 \\ 0 & \frac{T_{\mathrm{s}}}{\bar{L}_q} \end{bmatrix},$$

$$\bar{\mathbf{D}} = \begin{bmatrix} 0 \\ \frac{-T_{\mathrm{s}}}{\bar{L}_q}\bar{\psi}_f\omega_{\mathrm{e}}(k) \end{bmatrix}.$$

$\bar{L}_d$, $\bar{L}_q$, $\bar{R}_{\mathrm{s}}$ and $\bar{\psi}_f$ denote the mismatch inductances, resistance, and flux linkage, respectively, $\hat{i}_d$ and $\hat{i}_q$ are the estimated *dq*-axis current. Subtracting (20) from (22), the current error can be obtained as

$$\Delta\boldsymbol{i}(k+1) = \hat{\boldsymbol{i}}(k+1) - \boldsymbol{i}(k+1). \tag{24}$$

According to the parameter matrix, it can be written as

$$\Delta\boldsymbol{i}(k+1) = \Delta\mathbf{A}\boldsymbol{i}(k) + \Delta\mathbf{B}\boldsymbol{u}(k) + \Delta\mathbf{D} \tag{25}$$

where

$$\Delta\mathbf{A} = \begin{bmatrix} \left(\frac{R_{\mathrm{s}}}{L_d} - \frac{\bar{R}_{\mathrm{s}}}{\bar{L}_d}\right)T_{\mathrm{s}} & \left(\frac{\bar{L}_q}{\bar{L}_d} - \frac{L_q}{L_d}\right)\omega_{\mathrm{e}}(k)T_{\mathrm{s}} \\ -\left(\frac{\bar{L}_d}{\bar{L}_q} - \frac{L_d}{L_q}\right)\omega_{\mathrm{e}}(k)T_{\mathrm{s}} & \left(\frac{R_{\mathrm{s}}}{L_q} - \frac{\bar{R}_{\mathrm{s}}}{\bar{L}_q}\right)T_{\mathrm{s}} \end{bmatrix},$$

$$\Delta\mathbf{B} = \begin{bmatrix} \frac{T_{\mathrm{s}}}{\bar{L}_d} - \frac{T_{\mathrm{s}}}{L_d} & 0 \\ 0 & \frac{T_{\mathrm{s}}}{\bar{L}_q} - \frac{T_{\mathrm{s}}}{L_q} \end{bmatrix},$$

$$\Delta\mathbf{D} = \begin{bmatrix} 0 \\ \left(\frac{\psi_f}{L_q} - \frac{\bar{\psi}_f}{\bar{L}_q}\right)T_{\mathrm{s}}\omega_{\mathrm{e}}(k) \end{bmatrix}.$$

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

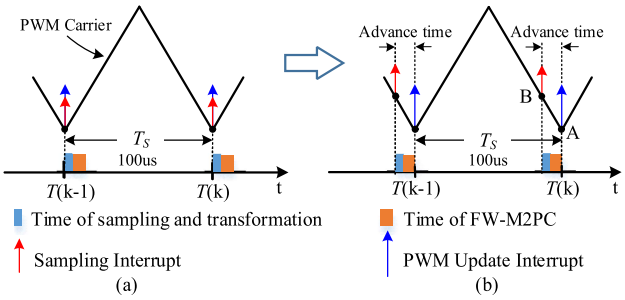


Fig. 9. Comparison of sampling and PWM update interrupt under two methods. (a) One-delay compensation. (b) Pre-triggering.

Similarly, the current error at $k+2$ instant can be described as

$$\Delta \boldsymbol{i}(k+2) = \underbrace{\Delta \mathbf{A}\boldsymbol{i}(k+1) + \Delta \mathbf{B}\boldsymbol{u}(k+1) + \Delta \mathbf{D}}_{\text{Part1}} + \underbrace{\bar{\mathbf{A}}\Delta \boldsymbol{i}(k+1)}_{\text{Part2}}. \tag{26}$$

From (26), the current error of one-delay compensation at $k+2$ instant can be divided into two parts: present error (Part1) and cumulative error (Part2). Part 1 and Part 2 are generated from the one-horizon and two-horizon predictions. To reduce the control delay and eliminate the cumulative error, pre-triggering sampling is applied to the proposed method. Fig. 9 presents the comparison of sampling and PWM update interrupt under one-delay compensation and pre-triggering methods.

Compared with the one-delay compensation, the sampling and calculation of the pre-triggering method are triggered in advance. In the experimental platform, the advance time is set to 7us, which is greater than the sum of the sampling time, the algorithm execution time, and the transmission time. The whole process starts with the sampling interruption and ends with the transmission completed, and the output is finally updated via the PWM interruption. Since the sampling period (T_s) is much greater than the advance time, it can be considered that the sampling values at point B and point A are consistent. This means that the switching states can be obtained and updated within the present period, that is, the control delay can be effectively reduced. Therefore, the cumulative error (Part 2) no longer exists since the prediction horizon of pre-triggering is reduced from two-horizon to one-horizon, which also improves the tolerance of parameter perturbation.

IV. Experimental Results

To verify the effectiveness of the proposed method, the comparative experiments are conducted on a PMSM test platform, as shown in Fig. 10. Parameters of the PMSM system are listed in Table I. The studied algorithm is implemented on Xilinx Zynq-7020, which integrates a feature-rich dual-core ARM Cortex-A9-based processing system (PS) and Xilinx programmable logic (PL) in a single device. The designed program diagram of PS and PL is shown in Fig. 11. The functions of PS include parameter initialization, the execution of the studied

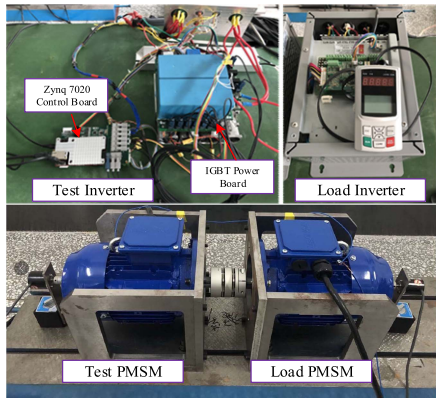


Fig. 10. Experimental test platform.

TABLE I
Parameters of the PMSM

Quantity	Value	Quantity	Value
DC Voltage	270 V	Stator Resistance	1.025 Ω
Rated Speed	1500 rpm	Rotor Flux Linkage	0.249 Wb
Rated Current	7.8 A	*d*-axis Inductance	9.10 mH
Rated Torque	12.5 N·m	*q*-axis Inductance	18.5 mH
Poles	3	Sampling period	100 μs

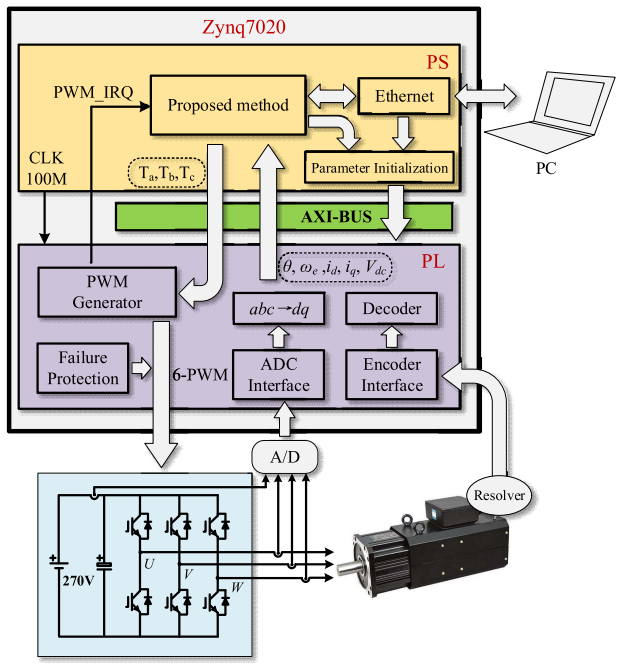


Fig. 11. Structure of program based on Zynq-7020.

algorithm, and ethernet communication with the host. In the part of PL, the user-defined IP cores are implemented with Verilog HDL, which includes analog signal acquisition and transformation, encoder signal acquisition and decoding, PWM output and PWM interrupt request (IRQ) generation, and fault protection.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

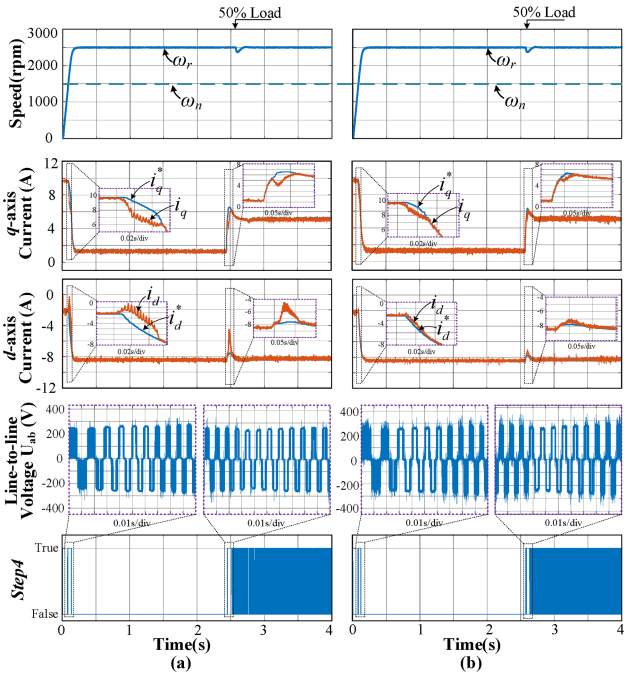


Fig. 12. Experimental results of conventional M2PC and proposed algebraical M2PC under acceleration and sudden load test. From top to bottom: Rotor speed, q-axis current, d-axis current, line-line voltage U_{ab}, and *Step 4*. (a) Conventional M2PC. (b) Proposed algebraical M2PC.

The experimental comparisons include the following parts: acceleration performance and load capacity comparison, dynamic performance comparison, comparison of one-delay compensation and pre-triggering, and comparison of the computational burden, where method1 and method2 refer to [18] with open-loop flux control method and voltage closed-loop based on field-oriented control [20], [22], respectively. In the proposed method, k_p and k_i of (19) are set to 0.5 and 350 in the experiment, respectively.

A. Acceleration Performance and Load Capacity Comparison

Fig. 12 presents the experimental results of conventional M2PC and proposed algebraical M2PC under acceleration and sudden load test. Since the difference between the conventional M2PC and the proposed algebraic M2PC is the duty cycle selection in the over-modulation region, the dq-axis current is amplified in the case of *Step4* = True (i.e., the over-modulation region), as shown in the zoom region. When *Step4* = True, the inverter enters into the over-modulation region, otherwise it is in the linear modulation region, and the line-line voltage U_{ab} waveform between the linear and over- modulated regions is shown in Fig. 12. From the comparison of the two methods, the dq-axis current of the conventional M2PC has difficulty following the reference dq-axis current and has a large tracking error during the over-modulation region. On the contrary, the proposed algebraical M2PC can effectively reduce the tracking error due to the duty cycle derived from the PCE minimization.

Figs. 13 and 14 present the experimental results of method1 and the proposed method in acceleration and sudden load-test,

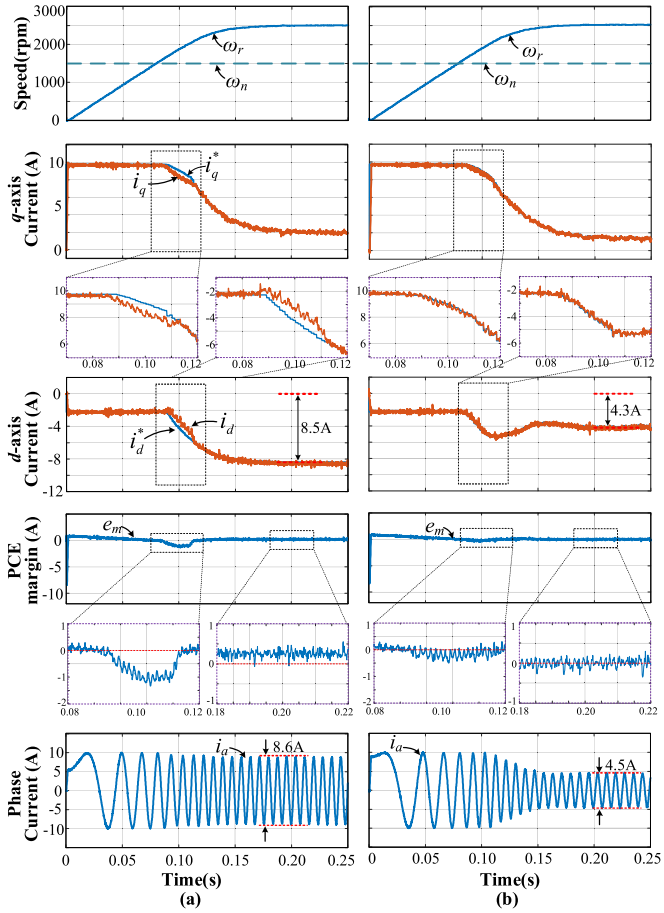


Fig. 13. Experimental results of method1 and proposed method under acceleration test. From top to bottom: rotor speed, q-axis current, d-axis current, PCE margin, and A-phase current. (a) Method1. (b) Proposed method.

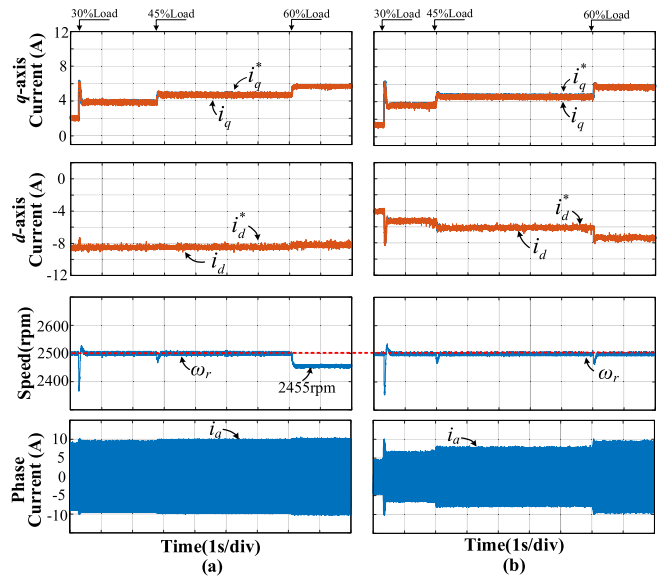


Fig. 14. Experimental results of method1 and proposed method under load capacity test. From top to bottom: q-axis current, d-axis current, rotor speed, and A-phase current. (a) Method1. (b) Proposed method.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

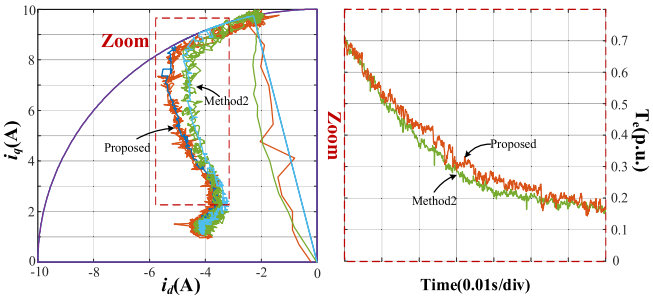


Fig. 15. Trajectories of *dq*-axis current and torque in acceleration.

respectively. From Fig. 13, it can be seen that the *dq*-axis tracking error of the proposed method is obviously less than that of method1 in the transition zone when the motor operation transits from the constant torque region to the FW Correspondingly, PCE margin of method1 inversely increases, but the PCE margin of the proposed method can be suppressed at zero in the transition zone. According to the mapping between the PCE and voltage, it indicates that the voltage of method1 is not enough for the extended speed during this interval. After arriving at the reference speed, the d-axis holding current of the proposed method reduces from −8.5 A to −4.3 A compared to the method1. The main reason for that is the inaccurate current distribution in method1. From the PCE margin point of view, the inverter in the proposed method is in quasi-saturation instead of unsaturation in method1. Then, 30%, 45%, and 60% load torque are sequentially applied to the load test, and the comparison results are shown in Fig. 14. With the increase of load torque, the rotor speed of method1 drops from 2500 rpm to 2455 rpm when the load torque suddenly changes from 45% to 60%, while the proposed method can effectively track the reference speed without error. It can be verified that the load capacity of the proposed method is better than that of method1, owing to the accurate current distribution of the proposed method.

B. Dynamic Performance Comparison

Figs. 15 and 16 are the experimental results of method2 and the proposed method in acceleration and sudden load change tests, respectively. From the trajectory comparison in Fig. 15, the generated torque of the proposed method is larger than that of method2 during the acceleration. Since the feedback of method2 is generated from the PI current controller, it is inevitably affected by the response rate, wind-up, and inaccuracy decouple problems during the saturation of current controller, which will cause inaccurate distribution of *dq*-axis current. On the contrary, the field-weakening current of the proposed method is generated through the PCE margin derived from the M2PC controller with PCE minimization. Compared to method 2, the *d*-axis current distribution is more accurate and responsive in the proposed method due to the fast prediction for inverter saturation and the enhanced decoupling performance.

Fig. 16 presents the responses with the 30%, 45%, and 60% sudden load torque under 2500 rpm. As the results show, the *dq*-axis current can rapidly follow the changing reference in the proposed method, but there are tracking errors in method2.

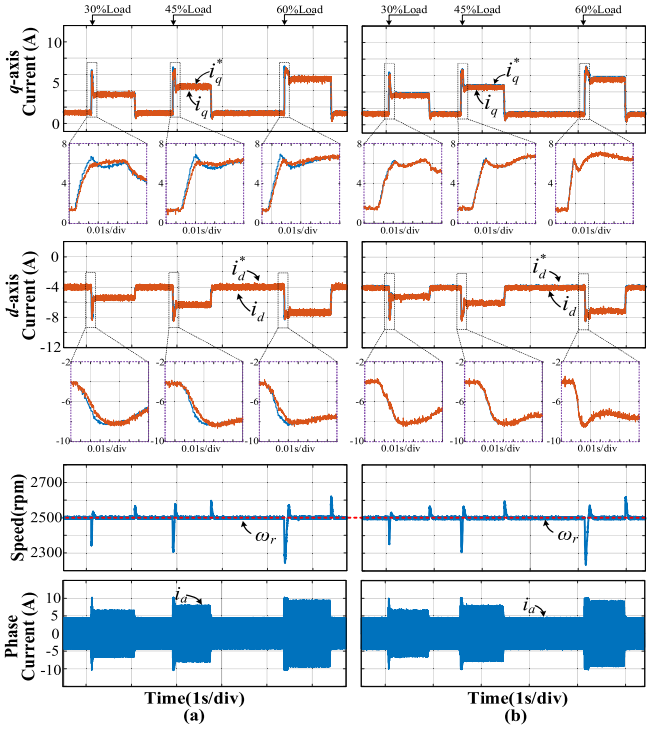


Fig. 16. Experimental results of method2 and proposed method under sudden load test. From top to bottom: *q*-axis current, *d*-axis current, rotor speed, and A-phase current. (a) Method2. (b) Proposed method.

Besides, both methods have consistent steady-state conditions. With the help of PCE margin as feedback, the fast and accurate distribution for *dq*-axis current can be obtained even under multiple constraints. It can be seen that the dynamic performance of the proposed method is much more satisfactory in FW region.

C. Comparison of One-Delay Compensation and Pre-Triggering

Fig. 17 shows the experimental results of the one-delay compensation method and pre-triggering method under the parameter perturbation. For a fair comparison, the implementation of the one-delay compensation method and pre-triggering method is based on the proposed PCE minimization (Part A of Section III) and PCE margin (Part B of Section III). Considering that the prediction equation under high-speed operation is more sensitive to inductance parameters, the inductances are used as the parameter perturbation and set to 75%, 100%, and 125% of rated value under 2500 rpm, respectively. It can be seen that the *dq*-axis current of both methods can accurately track the reference in different loads when the inductance is 100%. Nevertheless, when the inductance is 75% and 125%, the *dq*-axis current of one-delay compensation cannot track the reference. The main reason is that the cumulative error becomes larger as the inductance perturbation deteriorates. On the contrary, the pre-triggering can accurately track the reference, which also demonstrates the elimination of cumulative error.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

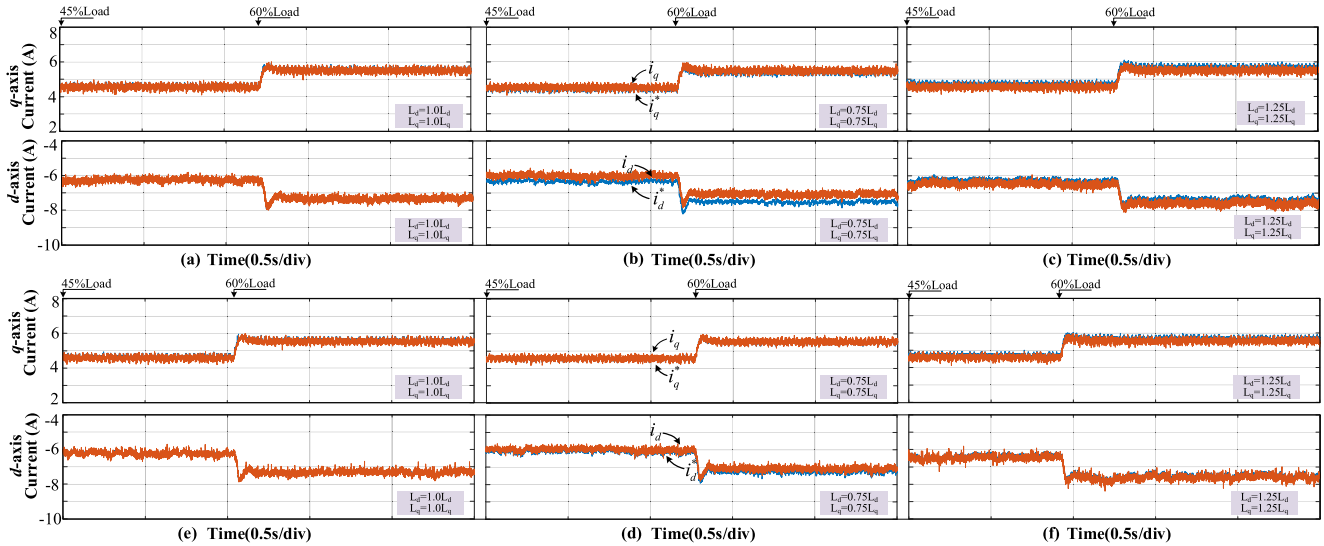


Fig. 17. Experimental results under parameter perturbation. (a), (b), (c) One-delay compensation method. (d), (e), (f) pre-triggering method.

TABLE II
COMPUTATIONAL BURDEN IN OVER-MODULATION

Method	Computing Process		Computing Time
	Algorithm	Number	
M2PC[18]	+, -	11	2.41μs
	*, /	16	
	sin, cos	2	
	asin, acos	2	
	sqrt	3	
Proposed M2PC	+, -	8	0.145μs
	*, /	5	

D. Comparison on Computational Burden

Table II compares the computational burden and the corresponding computing time of the proposed method and the method proposed in [18]. Note that for fairness, only algorithms for the over-modulation part are taken into account. The computing time of M2PC in [18] (2.41 μs) is 14 times that of the proposed M2PC (0.145 μs), meaning that the previous method will consume more resources to obtain the minimum PCE, which proves the algorithm's simplicity of the proposed method.

V. CONCLUSION

An integrated modulated-model-predictive-control scheme has been proposed to improve the dynamic performance and load capacity by minimizing tracking error, guiding accurate operation trajectory, and reducing control delay. The algebraical method achieves the minimized tracking error with concise calculation in the over-modulation region. The field-weakening method using PCE margin is designed and obtains accurate trajectory guidance under different working conditions. The pre-triggering mechanism reduces the control delay and eliminates the cumulative error, improving tolerance to parameter perturbances.

APPENDIX

In the appendix, the stability of the proposed methods is proved, and the specific process is as follows. According to (1), the dynamic model of the PMSM in dq-axis current can be further simplified as

$$\begin{aligned}\frac{\mathrm{d}i_d}{\mathrm{dt}} &= -m_1 i_d + m_2 u_d + m_3 \omega_e i_q \\ \frac{\mathrm{d}i_q}{\mathrm{dt}} &= -m_4 i_q + m_5 u_q - \frac{1}{m_3}\omega_e i_d - m_6 \omega_e \end{aligned} \tag{27}$$

where $m_1 = R_s/L_d$, $m_2 = 1/L_d$, $m_3 = L_q/L_d$, $m_4 = R_s/L_q$, $m_5 = 1/L_q$, $m_6 = \psi_f/L_d$.

From Fig. 2, the reference dq-axis current is yielded by the speed error and the predicted current error margin. Thus, the method can be simplified to a reference current tracking problem on the dq-axis. According to Lemma 1 in [33], two new state variables z_1 and z_2 can be presented, as

$$\begin{aligned} z_1 &= \frac{\mathrm{d}\eta_d}{\mathrm{dt}} + \varepsilon_d \eta_d, \eta_d = i_d^* - i_d \\ z_2 &= \frac{\mathrm{d}\eta_q}{\mathrm{dt}} + \varepsilon_q \eta_q, \eta_q = i_q^* - i_q \end{aligned} \tag{28}$$

where η_d and η_q are the error between the reference current and the feedback current in dq-axis, ε_d and ε_q are the decay rate of the dq-axis current error. By taking the derivatives of (28) for the new state variables z_1 and z_2, the new dynamic model for the new state variables can be obtained, as

$$\frac{\mathrm{d}z_1}{\mathrm{dt}} = m_2\left(u_d - \delta_d\right), \frac{\mathrm{d}z_2}{\mathrm{dt}} = m_5\left(u_q - \delta_q\right). \tag{29}$$

where the lumped quantities (δ_d, δ_q) are composed of constant parameter vectors $\mu d \in \mathrm{R}^{2\mathrm{x}1}$, $\mu q \in \mathrm{R}^{3\mathrm{x}1}$. The state-depended vectors x_d, x_q are written as

$$\begin{aligned} &\delta_d = \mu_d^T x_d, \delta_q = \mu_q^T x_q, \\ &x_d = \left[i_d \omega_e i_q\right]^T, x_q = \left[i_q \omega_e i_d \omega_e\right]^T. \end{aligned} \tag{30}$$

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

Substituting (30) into (29), the new dynamic model of PMSM is further described as

$$\frac{\mathrm{d}z_1}{\mathrm{dt}} = m_2\left(u_d - \mu_d^T x_d\right), \frac{\mathrm{d}z_2}{\mathrm{dt}} = m_5\left(u_q - \mu_q^T x_q\right). \quad (31)$$

From the implementation steps of the proposed M2PC, the duty cycle (d_a, d_b) of adjacent voltage vectors is calculated by the predictive current error (e_d, e_q). Thus, by reversely deriving the predictive current error, the reference dq-axis voltage can be presented as

$$\begin{bmatrix} u_d \\ u_q \end{bmatrix} = \begin{bmatrix} R_\mathrm{s} & -\omega_\mathrm{e} L_q \\ \omega_\mathrm{e} L_d & R_\mathrm{s} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} \frac{L_d}{T_\mathrm{s}} & 0 \\ 0 & \frac{L_q}{T_\mathrm{s}} \end{bmatrix} \begin{bmatrix} e_d \\ e_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_\mathrm{e}\psi_f \end{bmatrix} \quad (32)$$

where (e_d, e_q) can be presented from (3), as

$$\begin{bmatrix} e_d \\ e_q \end{bmatrix} = d_\mathrm{a} \begin{bmatrix} i_d^* - i_d|u_{\mathrm{a}d} \\ i_q^* - i_q|u_{\mathrm{a}q} \end{bmatrix} + d_\mathrm{b} \begin{bmatrix} i_d^* - i_d|u_{\mathrm{b}d} \\ i_q^* - i_q|u_{\mathrm{b}q} \end{bmatrix}. \quad (33)$$

Using Lemma 2 in [33], the feedback terms and compensating terms of the dynamic model (31) can be presented as

$$u_d = -k_d z_1 + \mu_d^T x_d, u_q = -k_q z_2 + \mu_q^T x_q \quad (34)$$

where $k_d > 0$ and $k_q > 0$ are the feedback gains. Combining the principle M2PC, (32) and (34), the feedback gain (k_d, k_q) of the actual control system can be obtained as

$$\begin{cases} k_d = L_d/T_\mathrm{s},\ k_q = L_q/T_\mathrm{s} & \text{if } d_\mathrm{a} + d_\mathrm{b} \le 1 \\ 0 < k_d < L_d/T_\mathrm{s},\ 0 < k_q < L_q/T_\mathrm{s} & \text{else} \end{cases}. \quad (35)$$

Note that, in the actual system, the feedback gain will be restricted in the case of $d_\mathrm{a} + d_\mathrm{b} > 1$ (i.e., in the over-modulation region). However, this restriction does not affect the sign of the feedback gain.

Substituting (34) into (31), the closed-loop dynamic of (31) with control laws (34) is obtained as

$$\frac{\mathrm{d}z_1}{\mathrm{dt}} = -m_2 k_d z_1, \frac{\mathrm{d}z_2}{\mathrm{dt}} = -m_5 k_q z_2. \quad (36)$$

A Lyapunov function can be designed as

$$V(t) = \frac{z_1^2}{2} + \frac{z_2^2}{2}. \quad (37)$$

From (37), it can be confirmed that the designed Lyapunov function is always positive. Then, the first-order derivative of (37) can be written as

$$\frac{\mathrm{d}V}{\mathrm{dt}} = -m_2 k_d z_1^2 - m_5 k_q z_2^2. \quad (38)$$

Due to the positive feedback gain $k_d > 0$ and $k_q > 0$ presented in (35), it inferred dV/dt < 0 guaranteeing the asymptotical stability.

REFERENCES

[1] W. Xu, M. M. Ismail, Y. Liu, and M. R. Islam, "Parameter optimization of adaptive flux-weakening strategy for permanent-magnet synchronous motor drives based on particle swarm algorithm," *IEEE Trans. Ind. Electron.*, vol. 34, no. 12, pp. 12128–12140, Dec. 2019.

[2] A. Balamurali, G. Feng, C. Lai, J. Tjong, and N. C. Kar, "Maximum efficiency control of PMSM drives considering system losses using gradient descent algorithm based on DC power measurement," *IEEE Trans. Energy Convers.*, vol. 33, no. 4, pp. 2240–2249, Dec. 2018.

[3] Q. Hou, S. Ding, and X. Yu, "Composite super-twisting sliding mode control design for PMSM speed regulation problem based on a novel disturbance observer," *IEEE Trans. Energy Convers.*, vol. 36, no. 4, pp. 2591–2599, Dec. 2021.

[4] S. Liu, C. Liu, Y. Huang, and H. Zhao, "Model predictive two-target current control for OW-PMSM," *IEEE Trans. Power Electron.*, vol. 36, no. 3, pp. 3224–3235, Mar. 2021.

[5] Z. Song, S. Hu, and Z. Bao, "Variable action period predictive flux control strategy for permanent magnet synchronous machines," *IEEE Trans. Power Electron.*, vol. 35, no. 6, pp. 6185–6197, Jun. 2020.

[6] J. Rodriguez et al., "Predictive current control of a voltage source inverter," *IEEE Trans. Ind. Electron.*, vol. 54, no. 1, pp. 495–503, Feb. 2007.

[7] X. Li, W. Tian, X. Gao, Q. Yang, and R. Kennel, "A generalized observer-based robust predictive current control strategy for PMSM drive system," *IEEE Trans. Ind. Electron.*, vol. 69, no. 2, pp. 1322–1332 , Feb. 2022.

[8] J. Gao, C. Gong, W. Li, and J. Liu, "Novel compensation strategy for calculation delay of finite control set model predictive current control in PMSM," *IEEE Trans. Ind. Electron.*, vol. 67, no. 7, pp. 5816–5819, Jul. 2020.

[9] P. Kakosimos and H. Abu-Rub, "Predictive speed control with short prediction horizon for permanent magnet synchronous motor drives," *IEEE Trans. Power Electron.*, vol. 33, no. 3, pp. 2740–2750, Mar. 2018.

[10] X. Zhang and Y. He, "Direct voltage-selection based model predictive direct speed control for PMSM drives without weighting factor," *IEEE Trans. Power Electron.*, vol. 34, no. 8, pp. 7838–7851, Aug. 2019.

[11] M. Tomlinson, H. D. T. Mouton, R. Kennel, and P. Stolze, "A fixed switching frequency scheme for finite-control-set model predictive control—concept and algorithm," *IEEE Trans. Ind. Electron.*, vol. 63, no. 12, pp. 7662–7670, Dec. 2016.

[12] Y. Zhang, Y. Bai, and H. Yang, "A universal multiple-vector-based model predictive control of induction motor drives," *IEEE Trans. Power Electron.*, vol. 33, no. 8, pp. 6957–6969, Aug. 2018.

[13] Z. Ma, S. Saeidi, and R. Kennel, "FPGA implementation of model predictive control with constant switching frequency for PMSM drives," *IEEE Trans. Ind. Inform.*, vol. 10, no. 4, pp. 2055–2063, Nov. 2014.

[14] D. Xiao, K. S. Alam, M. Norambuena, M. F. Rahman, and J. Rodriguez, "Modified modulated model predictive control strategy for a grid-connected converter," *IEEE Trans. Ind. Electron.*, vol. 68, no. 1, pp. 575–585, Jan. 2021.

[15] L. Tarisciotti, P. Zanchetta, A. Watson, J. C. Clare, M. Degano, and S. Bifaretti, "Modulated model predictive control for a three-phase active rectifier," *IEEE Trans. Ind. Appl.*, vol. 51, no. 2, pp. 1610–1620, Mar./Apr. 2015.

[16] F. Donoso, A. Mora, R. Cárdenas, A. Angulo, D. Sáez, and M. Rivera, "Finite-set model-predictive control strategies for a 3L-NPC inverter operating with fixed switching frequency," *IEEE Trans. Ind. Electron.*, vol. 65, no. 5, pp. 3954–3965, May 2018.

[17] E. Fuentes, C. A. Silva, and R. M. Kennel, "MPC implementation of a quasi-time-optimal speed control for a PMSM drive, with inner modulated-FS-MPC torque control," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3897–3905, Jun. 2016.

[18] C. F. Garcia, C. A. Silva, J. R. Rodriguez, P. Zanchetta, and S. A. Odhano, "Modulated model-predictive control with optimized overmodulation," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 7, no. 1, pp. 404–413, Mar. 2019.

[19] L. Sepulchre, M. Fadel, M. Pietrzak-David, and G. Porte, "MTPV flux-weakening strategy for PMSM high speed drive," *IEEE Trans. Ind. Appl.*, vol. 54, no. 6, pp. 6081–6089, Nov./Dec. 2018.

[20] T. Huber, W. Peters, and J. Bocker, "Voltage controller for flux weakening operation of interior permanent magnet synchronous motor in automotive traction applications," in *Proc. IEEE Int. Electric Mach. Drives Conf.*, 2015, pp. 1078–1083.

[21] X. Zhang, G. H. B. Foo, and M. F. Rahman, "A robust field-weakening approach for direct torque and flux controlled reluctance synchronous motors with extended constant power speed region," *IEEE Trans. Ind. Electron.*, vol. 67, no. 3, pp. 1813–1823, Mar. 2020.

[22] B. Wang, Y. Zhao, Y. Yu, G. Wang, D. Xu, and Z. Dong, "Speed-sensorless induction machine control in the field-weakening region using discrete speed-adaptive full-order observer," *IEEE Trans. Power Electron.*, vol. 31, no. 8, pp. 5759–5773, Aug. 2016.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.

[23] S.-Y. Jung, C. C. Mi, and K. Nam, "Torque control of IPMSM in the field-weakening region with improved DC-link voltage utilization," *IEEE Trans. Ind. Electron.*, vol. 62, no. 6, pp. 3380–3387, Jun. 2015.
[24] Z. Mynar, L. Vesely, and P. Vaclavek, "PMSM model predictive control with field-weakening implementation," *IEEE Trans. Ind. Electron.*, vol. 63, no. 8, pp. 5156–5166, Aug. 2016.
[25] M. Preindl and S. Bolognani, "Model predictive direct torque control with finite control set for PMSM drive systems, part 2: Field weakening operation," *IEEE Trans. Ind. Inform.*, vol. 9, no. 2, pp. 648–657, May 2013.
[26] P. Václavek and P. Blaha, "Field weakening in PMSM model based predictive control," in *Proc. IEEE Int. Conf. Power Energy*, 2010, pp. 330–335.
[27] J. Liu, C. Gong, Z. Han, and H. Yu, "IPMSM model predictive control in flux-weakening operation using an improved algorithm," *IEEE Trans. Ind. Electron.*, vol. 65, no. 12, pp. 9378–9387, Dec. 2018.
[28] Y. Zhang, Y. Bai, H. Yang, and B. Zhang, "Low switching frequency model predictive control of three-level inverter-fed IM drives with speed-sensorless and field-weakening operations," *IEEE Trans. Ind. Electron.*, vol. 66, no. 6, pp. 4262–4272, Jun. 2019.
[29] Y. Zhang, B. Zhang, H. Yang, M. Norambuena, and J. Rodriguez, "Generalized sequential model predictive control of IM drives with field-weakening ability," *IEEE Trans. Power Electron.*, vol. 34, no. 9, pp. 8944–8955, Sep. 2019.
[30] Z. Zheng and D. Sun, "Model predictive flux control with cost function-based field weakening strategy for permanent magnet synchronous motor," *IEEE Trans. Power Electron.*, vol. 35, no. 2, pp. 2151–2159, Feb. 2020.
[31] S. A. Odhano, P. Giangrande, R. I. Bojoi, and C. Gerada, "Self-commissioning of interior permanent- magnet synchronous motor drives with high-frequency current injection," *IEEE Trans. Ind. Appl.*, vol. 50, no. 5, pp. 3295–3303, Sep./Oct. 2014.
[32] P. Cortes, J. Rodriguez, C. Silva, and A. Flores, "Delay compensation in model predictive current control of a three-phase inverter," *IEEE Trans. Ind. Electron.*, vol. 59, no. 2, pp. 1323–1325, Feb. 2012.
[33] H. T. Nguyen and J. Jung, "Finite control set model predictive control to guarantee stability and robustness for surface-mounted PM synchronous motors," *IEEE Trans. Ind. Electron.*, vol. 65, no. 11, pp. 8510–8519, Nov. 2018.



Qinghua Dong was born in Shandong Province, China, in 1994. He received the B.S. degree in electrical engineering from the China University of Mining and Technology, Xuzhou, China, in 2016, and the M.S. degree in electrical engineering in 2018 from the Harbin Institute of Technology, Harbin, China, where he is currently working toward the Ph.D. degree in electrical engineering. His research interests include ac motor drives and model predictive control.



Bo Wang (Member, IEEE) was born in Shandong Province, China, in 1987. He received the B.S. degree in electrical engineering from Northwestern Polytechnical University, Xi'an, China, in 2011, and the M.S. and the Ph.D. degrees in electrical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2013 and 2017, respectively. In 2017, he joined the School of Electrical Engineering and Automation, HIT, where he is currently an Associate Professor of electrical engineering. He has authored more than 40 technical papers published in journals and conference proceedings. His research interests include ac motor drives, sensorless control, and nonlinear control theory.



Liqun Xia was born in May 1968. He received the Graduate degree in fluid transmission and control from the Department of Fluid Transmission and Control, Beijing Institute of Technology, Beijing, China, in 1990, and the master's degree in aviation electronic engineering and the Doctorate degree in navigation guidance and control from Northwestern Polytechnical University, Xi'an, China, in 2001 and 2009, respectively. He is currently an Actuation System Chief Engineer with FACRI, Xian, China. His research focuses on hydraulics.



Yong Yu was born in Jilin Province, China, in 1974. He received the B.S. degree in electromagnetic measurement and instrumentation, and the M.S. and Ph.D. degrees in electrical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 1995, 1997, and 2003, respectively. From 2004 to 2014, he was an Associate Professor with the Department of Electrical Engineering, HIT, where he has been a Professor of Electrical Engineering since 2014. His research interests include AC motor drives, power quality mitigation, and fault diagnosis and tolerant control of inverter.



Minghe Tian was born in Henan Province, China, in 1996. He received the B.S. degree in electrical engineering from Northeastern University, Shenyang, China, in 2018, and the M.S. degree in electrical engineering in 2020 from the Harbin Institute of Technology, Harbin, China, where he is currently working toward the Ph.D. degree in electrical engineering. His research interests include PMSM drives and anti-disturbance control theory.



Dianguo Xu (Fellow, IEEE) received the B.S. degree in control engineering from Harbin Engineering University, Harbin, China, in 1982, and the M.S. and the Ph.D. degrees in electrical engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 1984 and 1989, respectively. In 1984, he was an Assistant Professor with the Department of Electrical Engineering, HIT, where he has been a Professor with the Department of Electrical Engineering, since 1994. From 2000 to 2010, he was the Dean of School of Electrical Engineering and Automation, HIT. From 2014 to 2020, he was the Vice President with HIT. He has authored or coauthored more than 600 technical papers. His research interests include renewable energy generation technology, power quality mitigation, sensorless vector-controlled motor drives, and high performance PMSM servo system. Prof. Xu is the co-EIC of IEEE TRANSACTIONS ON POWER ELECTRONICS, and an Associate editor of IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS. He is also the Chairman of IEEE Harbin Section.

Authorized licensed use limited to: Izmir Katip Celebi Univ. Downloaded on October 28,2026 at 15:58:40 UTC from IEEE Xplore. Restrictions apply.